

BANKING TRANSACTION ANALYSIS

1. Project Overview

In the highly competitive landscape of the financial services industry, understanding customer behavior and optimizing operational efficiency is paramount. This project involves the development of a Business Intelligence (BI) Dashboard designed to transform raw retail banking transaction data into actionable insights. The system empowers management to monitor omnichannel transaction performance, evaluate customer loyalty, and detect financial risks, thereby driving data-driven decision-making.

2. Business Objectives

- **Customer Insights:** Analyze customer segment migration and monitor the overall health of the customer base.
- **Revenue & Product Performance:** Evaluate core revenue streams, with a focus on the Fee Burden across various financial products.
- **Retention Tracking:** Measure Customer Lifetime Value (CLV) and monthly retention rates (via Cohort Analysis).
- **Risk Management:** Monitor behavioral anomalies and potential credit risks.

3. Tech Stack & Skills Used

- **Tools:** Power BI (Data Visualization), Power Query (Data Transformation).
- **Skills:** Advanced DAX (Data Analysis Expressions), Data Modeling, Dashboard UI/UX Design, Data Storytelling.
- **Analytical Models:** RFM (Recency, Frequency, Monetary), Cohort Analysis, Pareto Principle.

4. Dashboard Architecture

The dashboard is structured into 5 main pages, providing a comprehensive view from high-level summaries to granular details:

- **Introduction:** Summarizes the business context, core capabilities, and top analytical findings.
- **Business Overview & Customer Behavior:** Analyzes demographics, income impacts, and the geographical distribution of transaction values (utilizing Decomposition Trees & Map Visuals).
- **RFM Analysis:** Evaluates dynamic customer segmentation based on RFM metrics to identify high-value "Champions" and churn-prone "At-Risk" groups.
- **Revenue & Product Analysis:** Tracks omnichannel transaction flows and assesses product-specific fee burdens (e.g., Loans, Credit Cards) across different income segments.
- **Cohort Analysis:** Visualizes the customer lifecycle, acquisition trends, and monthly Retention Rates.

5. Key Findings

- **Primary Revenue Drivers:** The *Middle Income* segment and *Loan* products are the core engines generating the highest transaction volume and fee revenue.
- **Retention Crisis (2025):** A sharp decline in New Customers (down 61.6% in 2025), a Retention Rate drop to 30.8%, and a declining CLV highlight critical bottlenecks in acquisition and long-term user engagement.
- **Value Segmentation (RFM Focus):** The Pareto principle holds strong, with the top 3 segments (Champions, Loyal, At Risk) consistently contributing 65% of the

total monetary value. The *At Risk* group generates high revenue but poses a significant churn threat, requiring immediate attention.

- **Fee Burden Inequity:** The *Low Income* segment faces the highest fee burden ratio (1.87% compared to 0.41% for the high-income group), posing significant affordability issues and elevating churn risk.
- **Omnichannel & Risk:** Transactions are evenly distributed across all channels (ATM, Branch, Mobile, Online). While no direct fraud evidence is visible, late payments and penalty fees from the *At Risk* group warrant close monitoring.

6. Strategic Recommendations

6.1. Overhaul Acquisition and Onboarding Strategies (Addressing the 2025 Crisis)

The severe 61.6% drop in new user acquisition and the steep decline in retention rate require immediate marketing interventions:

- **Root-cause Analysis:** Conduct a deep dive into 2025 marketing campaign data and channel performance to identify why the influx of new users halted compared to 2023/2024.
- **Implement a 90-Day Onboarding Journey:** Since the Cohort Analysis shows a massive drop-off from Month 0 to Month 1, introduce a structured onboarding sequence. Offer targeted incentives (e.g., cashback on the first 3 transactions or waived fees for the first month) to build habit-forming transaction behaviors early.
- **Re-activation Campaigns:** Deploy personalized email/push notification campaigns targeting the "Hibernating" and "About to Sleep" segments with low-barrier offers to bring them back to the active pool.

6.2. Alleviate the Low-Income Fee Burden

The disproportionate fee burden (1.87%) on the Low-Income segment is a critical churn trigger that must be managed to maintain market share:

- **Introduce Tiered Fee Structures:** Design "Maintenance-Free" or "Low-Fee" digital-only account tiers. By encouraging Low-Income customers to use digital

channels (Mobile/Online) instead of Branch services, the bank can reduce operational costs while passing the savings back to the customer.

- Proactive Payment Alerts: Implement an automated SMS/App notification system to remind customers of upcoming loan or credit card payment deadlines. This helps reduce late penalty fees, thereby improving customer satisfaction and financial health while lowering default risks.

6.3. Maximize Value from 'Champions' and 'Loyal' Segments

With the top segments generating 65% of total value, sustaining their engagement is crucial, especially since their average transaction frequency dropped slightly from 1.6 to 1.2:

- Hyper-Personalized Cross-Selling: Leverage their transaction history to offer pre-approved, customized Loan and Credit Card products—the highest revenue drivers.
- VIP Loyalty Programs: Introduce exclusive benefits for the "Champions" segment, such as priority customer service, premium credit card upgrades, or better interest rates, to incentivize a higher frequency of transactions and increase Customer Lifetime Value (CLV).

6.4. Mitigate Risk in High-Value 'At-Risk' Customers

The "At-Risk" segment contributes significantly to monetary value but exhibits a high likelihood of churning, alongside potential anomaly signals in loan payments:

- Early Warning System: Develop a churn-prediction model to flag "At-Risk" customers who show sudden drops in transaction frequency or consistent late payments.
- Dedicated Account Management: Assign branch managers or relationship managers to reach out to high-value "At-Risk" clients. Offering them debt-restructuring options or customized retention packages can save highly profitable relationships before they switch to competitors.